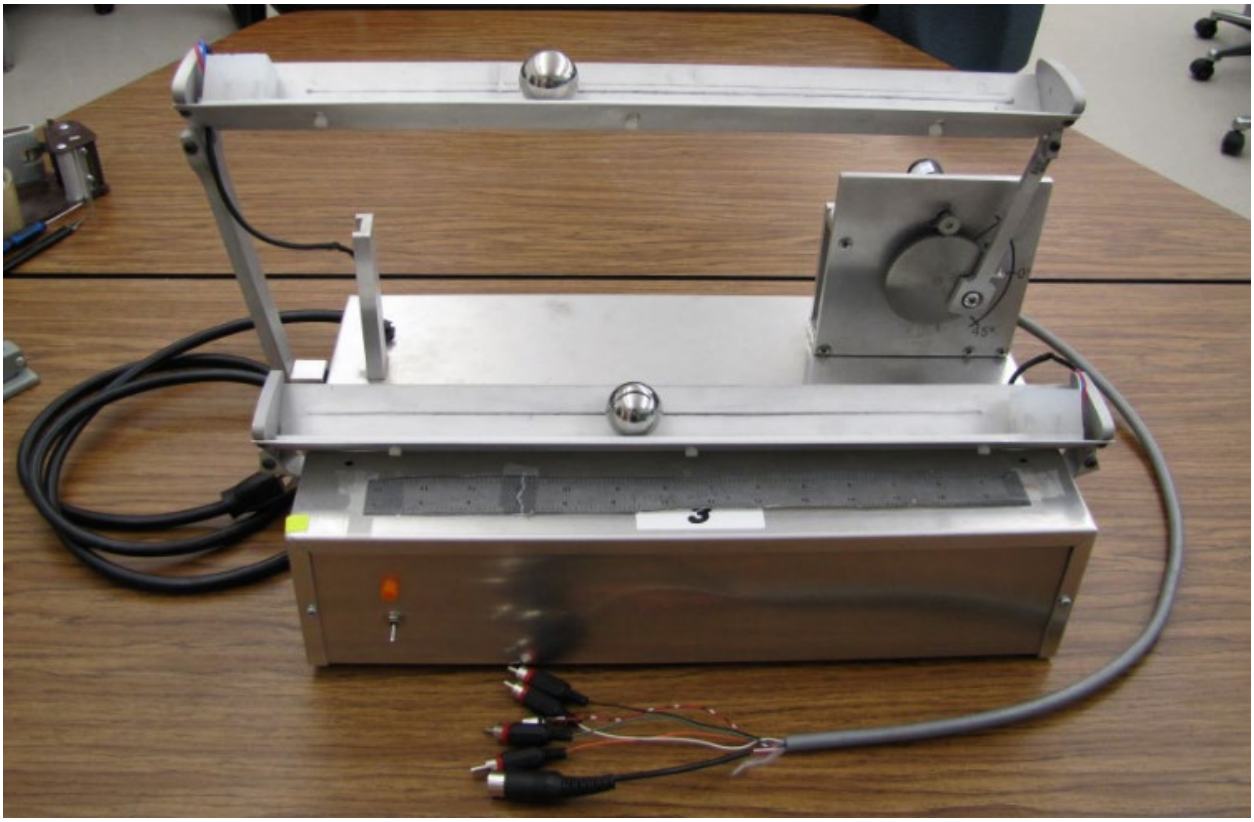
Ball and Beam: Angular Position Control using RSLogix 5000



Objectives:

- Control the angle of the motor on the Ball and Beam experiment using RSLogix 5000 and CompactLogix
- Observe the effects of P, PI, PD, and PID compensation.
- Observe the effects of a lowpass filter

Grading Rubric

- Working system (60 pts)
    - Rise time, $t_r \leq 0.5\ s$
    - Percent overshoot, $M_p \leq 5\%$
    - Settling time, $t_s \leq 0.7\ s$
    - Steady-state error, $e_{ss} \leq 3^{\circ}$
- Documented RS Logix 5000 program (15 pts)
    - Functional Block Diagram Program
    - Properties / setup of blocks added
        - Low pass filters (Trends showing tuned response)
        - Scale blocks

        - Etc.
    - Tags used and their definition
    - Setup of tasks (periodic vs. continuous, execution rate) and I/O modules (sample period)
- Document PID controller (15 pts)
    - Trends showing response to various gains
    - Best response: Gains used, response values (t_r, t_s, M_p, e_{ss})
- Conclusion (10 pts)

Procedure

The following is a sample Functional Block Diagram that you can used as you build your controller.

Document how you setup and used RSLogix 5000 so it can be a reference for you in the future.

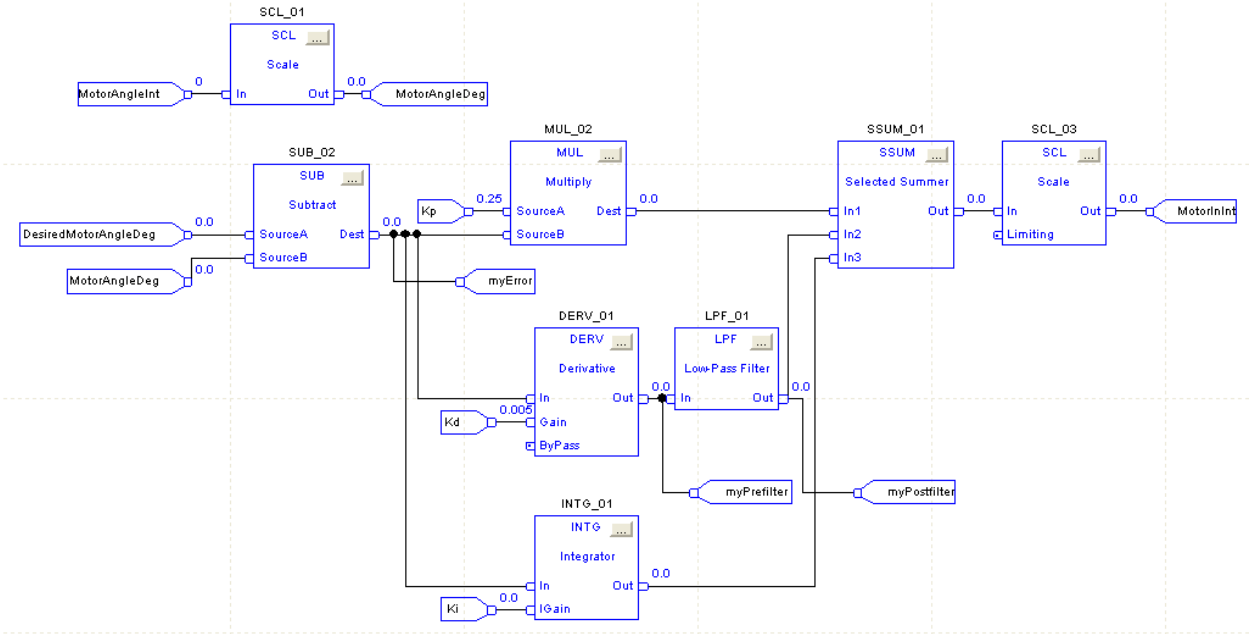
Use the trend feature to plot the response of your system. Also, use trend to tune and show the effect of the lowpass filter.

There are three options for obtaining a desired angle:

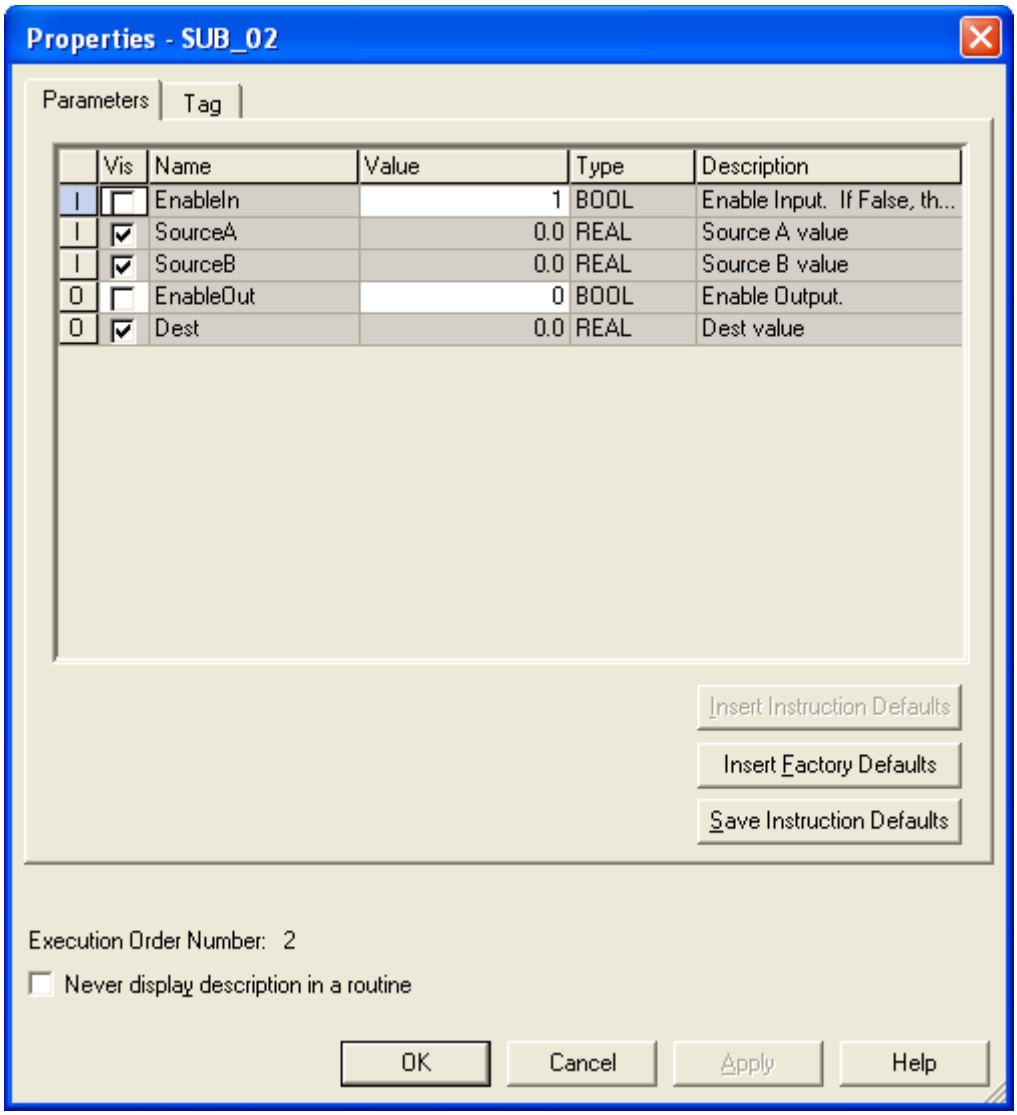
1. Use the tag monitor to enter the desired angle.
2. Use the switches (1, 2) to select desired an angular position: 00 = 0°, 10 = 30°, and 01= -30°.
3. Use the bottom ball position: (-5 V = -45°) to ($+5$ V = $+45^{\circ}$).

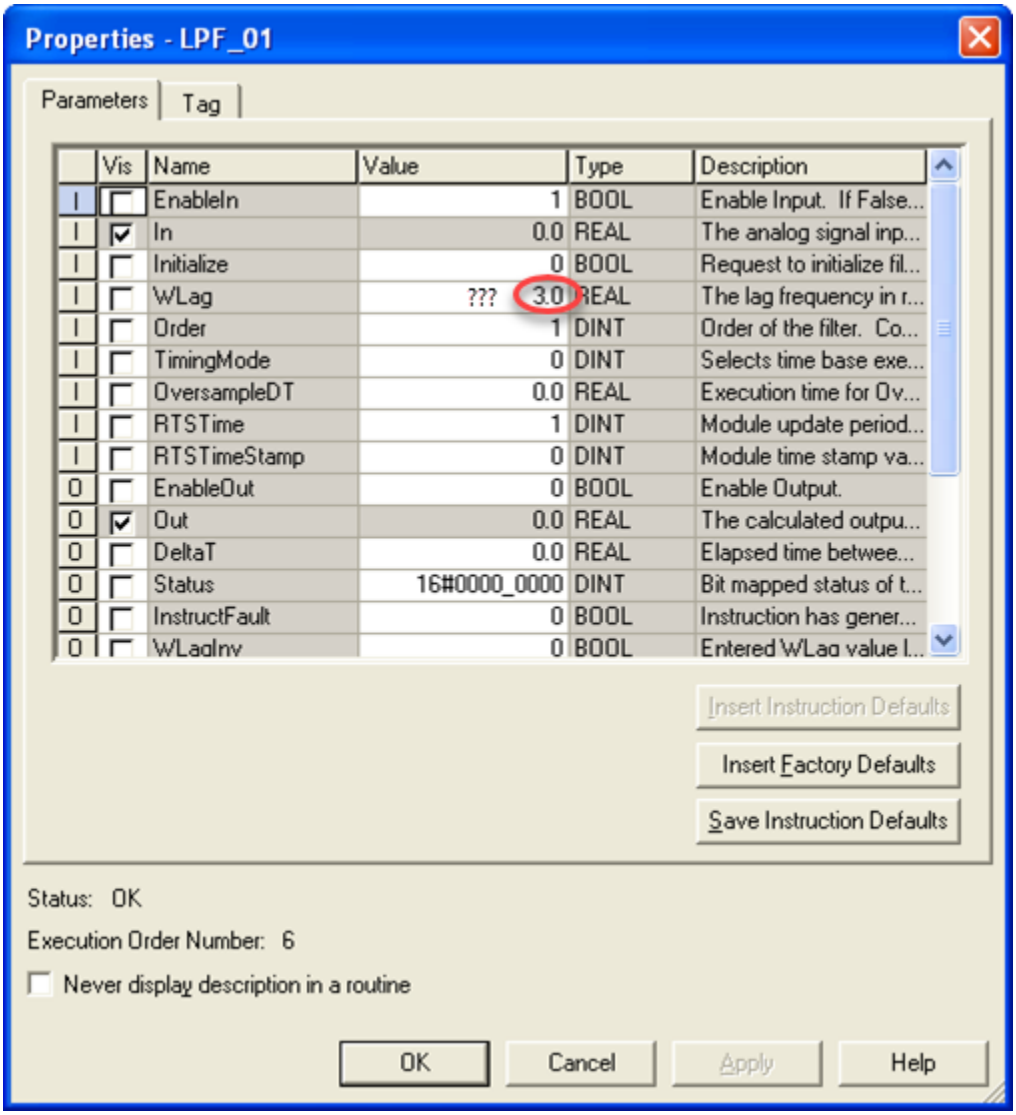
Any of these three options can be used.

Sample PLC Controller



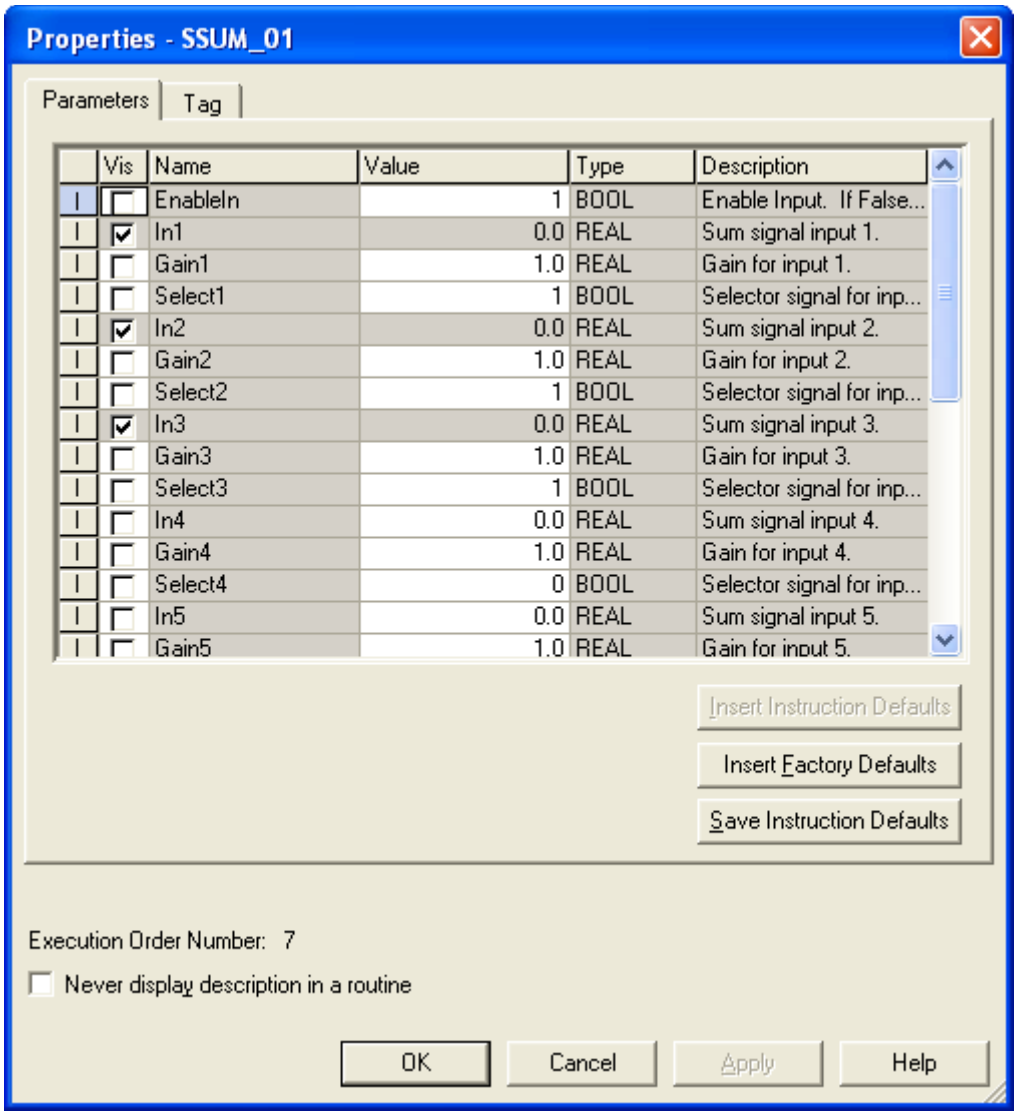
It works best to filter (using an LPF) the MotorAngleDeg signal following the scale function block SCL_01 (not shown above).

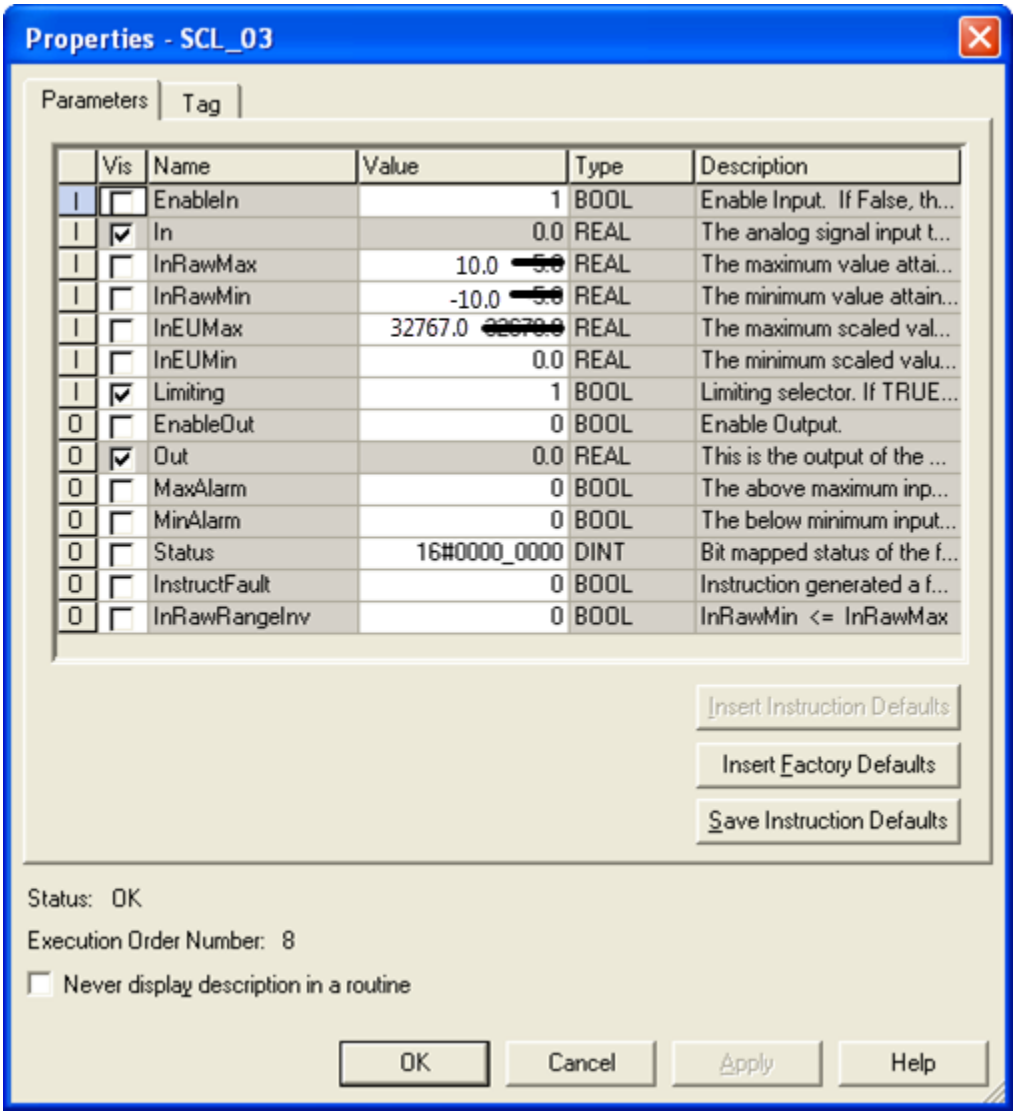

Properties - SUB_02
Parameters
Tag
Vis
Name
Value
Type
Description
I
EnableIn
1
BOOL
Enable Input. If False, th...
I
SourceA
0.0
REAL
Source A value
I
SourceB
0.0
REAL
Source B value
O
EnableOut
0
BOOL
Enable Output.
O
Dest
0.0
REAL
Dest value
Insert Instruction Defaults
Insert Factory Defaults
Save Instruction Defaults
Execution Order Number: 2
Never display description in a routine
OK
Cancel
Apply
Help



Trend LPF_01.in and LPF_01.out to determine the best value for WLag.

Also, add a Low-Pass Filter after SCL_01 (not shown) to filter the sensor data. Also, trend the input and output of this sensor to determine the optimal value for its WLag value.


Properties - SSUM_01
Parameters
Tag
Vis
Name
Value
Type
Description
EnableIn 1 BOOL Enable Input. If False...
In1 0.0 REAL Sum signal input 1.
Gain1 1.0 REAL Gain for input 1.
Select1 1 BOOL Selector signal for inp...
In2 0.0 REAL Sum signal input 2.
Gain2 1.0 REAL Gain for input 2.
Select2 1 BOOL Selector signal for inp...
In3 0.0 REAL Sum signal input 3.
Gain3 1.0 REAL Gain for input 3.
Select3 1 BOOL Selector signal for inp...
In4 0.0 REAL Sum signal input 4.
Gain4 1.0 REAL Gain for input 4.
Select4 0 BOOL Selector signal for inp...
In5 0.0 REAL Sum signal input 5.
Gain5 1.0 REAL Gain for input 5.
Insert Instruction Defaults
Insert Factory Defaults
Save Instruction Defaults
Execution Order Number: 7
Never display description in a routine
OK
Cancel
Apply
Help



What is the units of the input signal (SLC_03.in) for SLC_03?

Check list. Make sure:

1. The Main Task is configured to run in the Periodic mode at 5 ms.
2. The analog inputs and outputs are being updated at a rate of 5 ms.
3. Trends are updated every 5 ms.
4. Make sure you are using tags with a data type of REAL (and not INT) for process variables.

Include trends / plots that show:

1. Inputs and outputs of the tuned filters (show effects of the corner frequency, WLag)
2. The step response of the Desired and Actual angular positions of the motor:
    a. Proportional Control (show effects of Kp)
    b. Proportional and Derivative Control (show effects of Kd)

c. Proportional, Derivative, and Integral Control (show effects of Ki)

Appendix



Figure 6 BnB Wiring

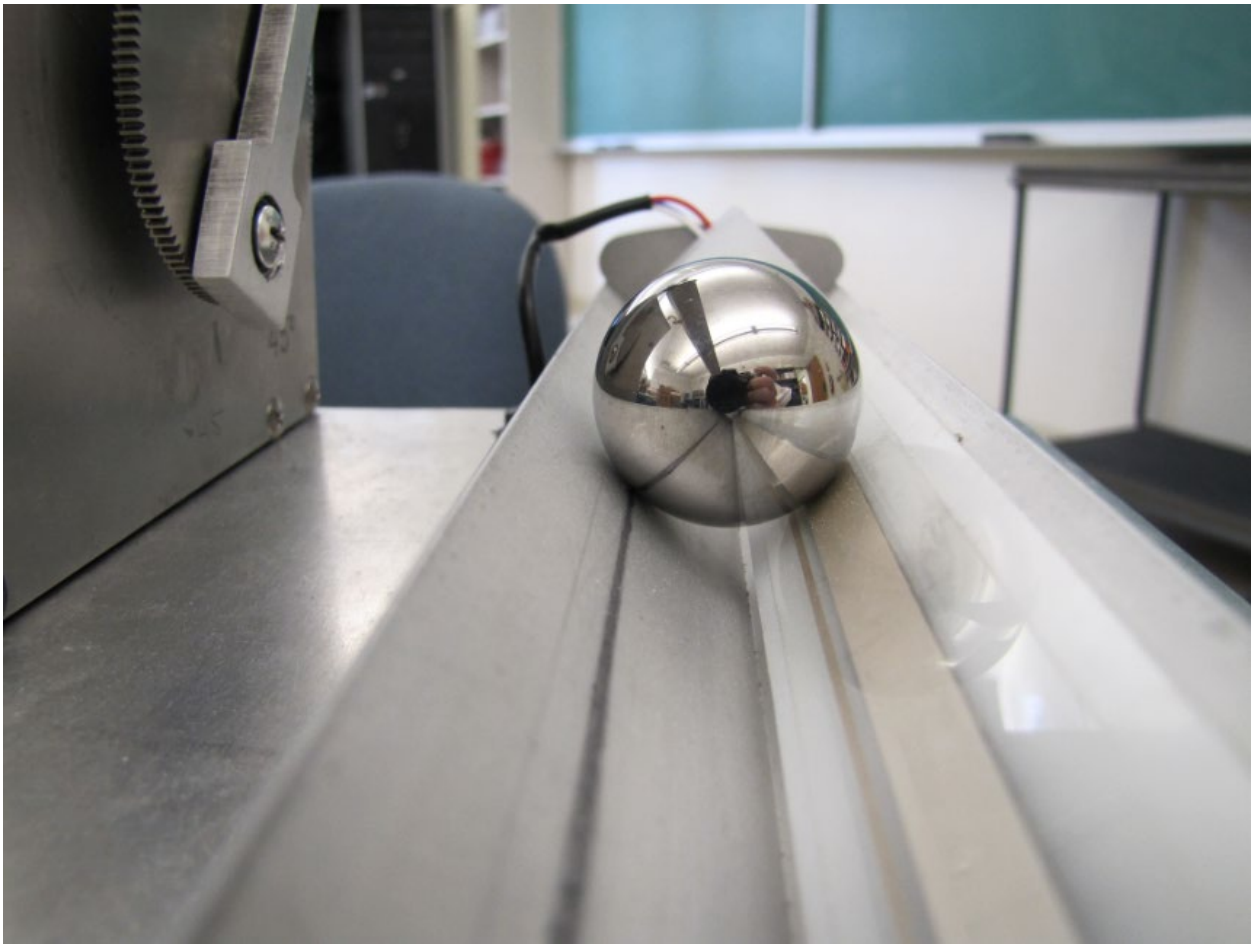


Figure 7 Alignment of position sensor strip so that it is tangent to the center of the ball.